

Two hackathons, one local vision demo

Cosmos3-Edge on an 8 GB Jetson Orin Nano

What worked, what the user changed, and what the numbers actually measure

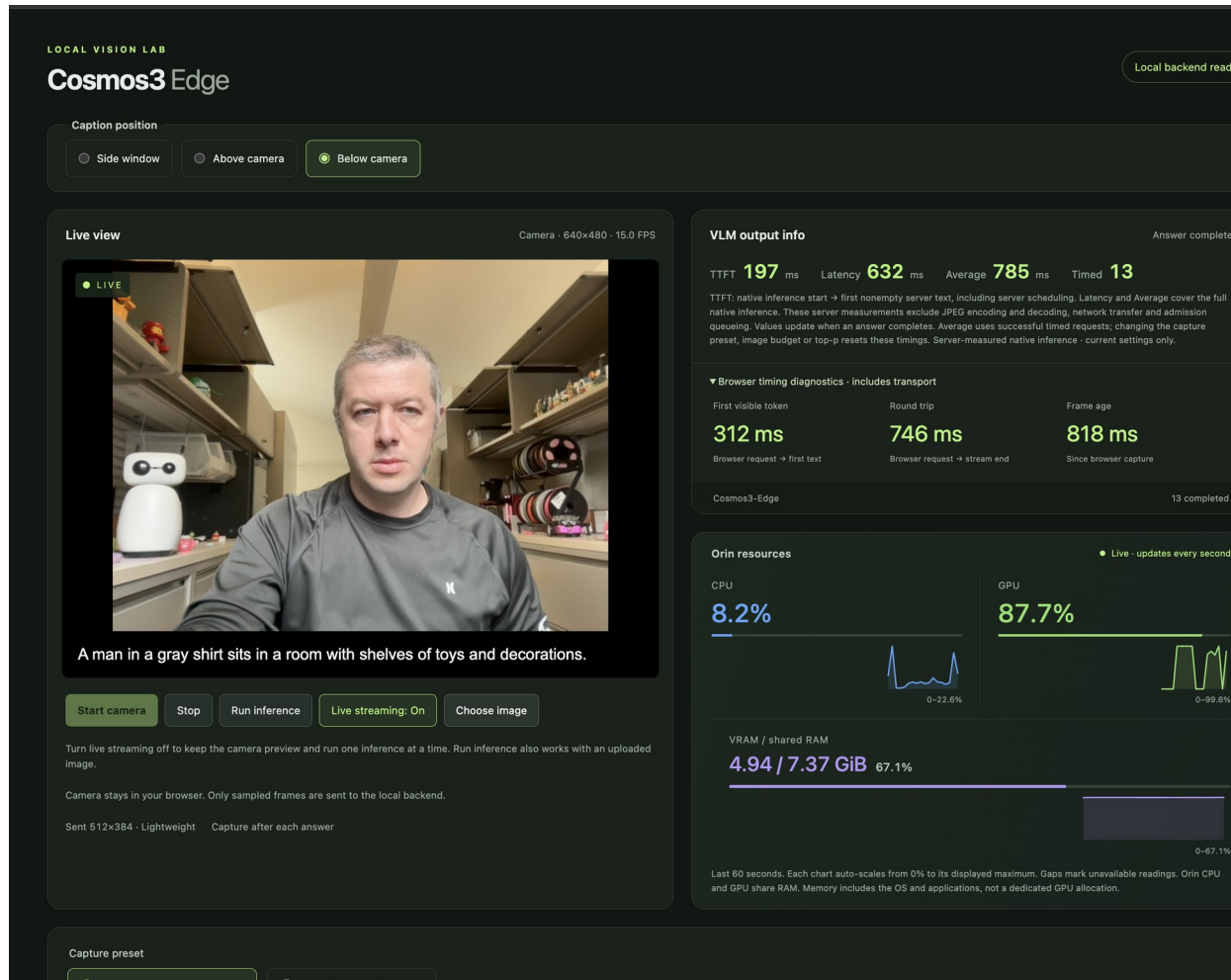
Two routes to useful edge vision

A local camera app within a small shared-memory budget

Design choice	First campaign	Current independent task
INT4 scope	169 text-layer linears	56 MLP linears
Power policy	MAXN_SUPER, fixed clocks	Stock 25 W, dynamic clocks
Application	Live VLM WebUI + JSON shim	Original UI + streamed text
Engineering focus	Resident runtime, broad quantization	Bring-up, streaming, audited timing

Different precision, power and serving choices require different performance claims.

One frame, two timing boundaries



197 ms

Server first text

312 ms

Browser first text

One displayed observation.
The gap is not a Wi-Fi test.

Human steering changed the defaults

Human request	What changed
“Compare fairly” to Live VLM	Prompt, output ceiling and capture preset aligned
Separate image and answer limits	Input budget 320 / 512 / Custom; output cap separate
Remove JPEG and network time	Native server timers; browser diagnostics retained
Restore the first demo	Original lightweight controls; temporary FP16 engine
Measure before switching back	Matched TTFT test, then final MLP INT4 deployment

Final: MLP INT4 with the first demo’s lightweight controls. Exact settings are in the appendix.

Matched time to first text

Same image and prompt. Same controls. Native server boundary.

Engine	Median	p95	Measured
Original FP16	249.5 ms	250.6 ms	30
MLP INT4	239.3 ms	240.7 ms	30

10.2 ms lower median for MLP on this fixture

Five warmups each · 256 visual tokens · cache off · 25 W / dynamic clocks
Sequential runs. One additional logged request started during the MLP run.

Historical gains use different workloads

First campaign, as reported in the supplied write-up

Reported change	Result	What it measures
Decode rate	22.5 → 76 tokens/s	Generated-token throughput
Process memory	6.02 → 3.70 GB	Process RSS, not dedicated VRAM
Later repeated-image run	253 ms median	Complete HTTP answer; 14 median tokens

253 ms to finish an answer and 239 ms to start text are different measurements.

Precision scope, power, input detail and cache behavior also differ. No cross-campaign speed ratio.

Stopping rules bounded the search

Trial in this task	Measured change	Decision
Earlier FP16 sampler	0.48% lower answer latency	Rejected; restored runtime
Compact MLP profile	1.19% lower answer latency 163.45 MiB less peak shared RAM	Stopped search; kept smaller profile

MLP failed the original 10% acceptance rule.

The agent clarified the rule after measurement: stop pursuing gains below 10%, while retaining the memory benefit.

Known color error remains. Neither global optimality nor general accuracy was established.

Working demo, explicit tradeoffs

Public NVIDIA stack

Cosmos3-Edge, JetPack and TensorRT-Edge-LLM provide the model and execution foundation.

Task-authored integration

Streaming camera UI, native timing, runtime controls, bounded tuning and evidence.

Live demo on the Orin

<https://192.168.6.252:8443/>

Architecture and runtime differences

Axis	First campaign source	Current final deployment
INT4 layers	169 attention + MLP + head	56 MLP only
Quantizer / plugin	RTN + MSE clipping / V2	Symmetric RTN / V1 + N4
Power / clocks	MAXN_SUPER / static	Stock 25 W / dynamic
Image token cap	320	512
Encoder cache	256 MiB native default ¹	0
Input / KV capacity	1536 / 2048	1024 / 1664
UI response	Whole-answer JSON	Incremental SSE text

¹ Historical executed cache state is not independently proven. Both retain FP16 components.

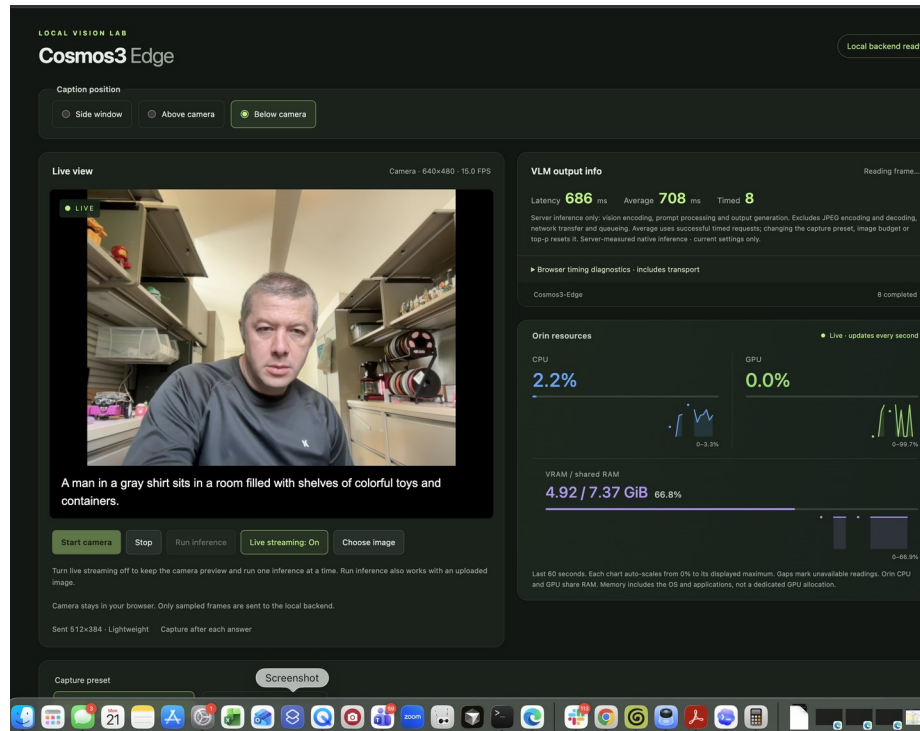
Recorded human and agent interventions

Stage	Intervention and consequence
Hardware bring-up	User recovery-jumper / connection steps; official USB-C flash
Initial assumptions	Agent chose lightweight capture, 64 output tokens, greedy decoding
Fair comparison and UI	Live VLM preset; HTTPS, prompts, graphs, caption controls
Freeze and bounded search	User froze a checkpoint and required 10% continuation gating
Policy correction	Agent kept smaller MLP profile after original acceptance failed
Runtime / timer steering	320 input cap + cache/clocks trial; then JPEG/network excluded
Rollback and measurement	Stop active sweep; restore lightweight, test FP16, then return to MLP

No measured labor-hours or intervention count is available for the first campaign.

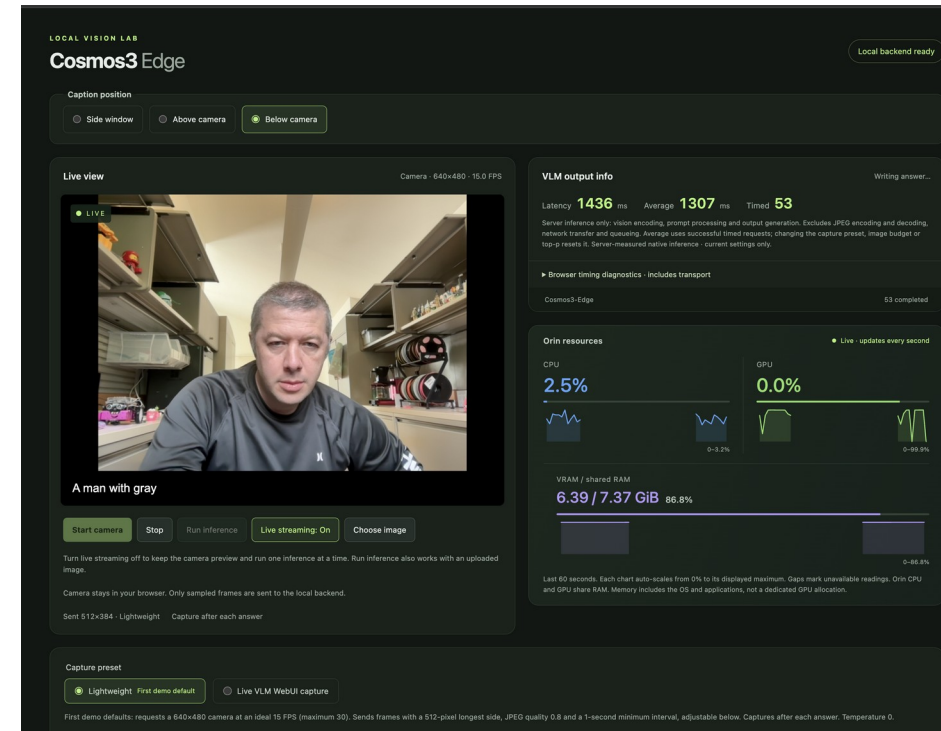
User screenshots are observations

MLP: user's "Astra Settings" label



686 ms latency · 708 ms average · n = 8
4.92 / 7.37 GiB shared RAM

Original FP16: user's "BF16" label



1,436 ms latency · 1,307 ms average · n = 53
6.39 / 7.37 GiB shared RAM

All show Lightweight capture. Advanced settings are outside the crop. These are not matched tests.

Keep each timing boundary explicit

Median on the fixed fixture	FP16	MLP INT4
Server first text	249.5 ms	239.3 ms
Loopback HTTP first text	267.7 ms	261.7 ms
Native complete answer	1,804.6 ms	1,009.0 ms
Actual output tokens	37	31

Different output lengths prevent a whole-answer engine-speed ratio.

The interrupted token-length sweep produced no finished fixed-ms / marginal-ms-per-token fit. Historical first-demo HTTP median: 265.54 ms. It uses the broader HTTP boundary.

Final defaults and the intermediate trial

Control	Intermediate request	Final first-demo settings
Input image-token cap	320; 512 and Custom options	512; 320 and Custom remain
Maximum output tokens	512, Live VLM reference	64
Temperature / top-p	0.7 / 0.95	0 / 1 (greedy)
Encoder cache / clocks	256 MiB / static enabled	0 / dynamic, stock 25 W
Capture / transmitted side	Live VLM capture preset	640×480 at 15 FPS / 512 px
JPEG / minimum interval	Capture-preset dependent	0.8 / 1 second

Input tokens control visual detail. Output tokens cap answer length. They are independent.

Attribution and reproducible evidence

NVIDIA

Model, JetPack and TensorRT-Edge-LLM. Existing encoder, runtime and kernels.

Live VLM WebUI and public contributors

Workflow and prompts. Apache-2.0; public contributors include tokk-nv and chitoku.

Campaign-specific work

Resident serving and broader INT4 in the first campaign. Independent bring-up, streaming UI, runtime controls and measurement in this task.

Pinned source: first 1375886 · current 935ea488 (cosmos3-edge-orin/)

Reports, request logs, image hashes and full source links are in the speaker notes and repository.